

A novel deep learning identifier for promoters and their strength using heterogeneous features

Aqsa Amjad^a, Saeed Ahmed^a, Muhammad Kabir^{a,*}, Muhammad Arif^{b,*}, Tanvir Alam^{b,*}

^a School of Systems and Technology, University of Management and Technology, Lahore 54770, Pakistan

^b College of Science and Engineering, Hamad Bin Khalifa University, Doha 34110, Qatar

ARTICLE INFO

Keywords:

Computational intelligence
Bioinformatics
Promoters
Convolutional neural network
Bidirectional long short-term memory

ABSTRACT

Promoters, which are short (50–1500 base-pair) in DNA regions, have emerged to play a critical role in the regulation of gene transcription. Numerous dangerous diseases, likewise cancer, cardiovascular, and inflammatory bowel diseases, are caused by genetic variations in promoters. Consequently, the correct identification and characterization of promoters are significant for the discovery of drugs. However, experimental approaches to recognizing promoters and their strengths are challenging in terms of cost, time, and resources. Therefore, computational techniques are highly desirable for the correct characterization of promoters from unannotated genomic data. Here, we designed a powerful bi-layer deep-learning based predictor named “PROCABLES”, which discriminates DNA samples as promoters in the first-phase and strong or weak promoters in the second-phase respectively. The proposed method utilizes five distinct features, such as word2vec, k-spaced nucleotide pairs, trinucleotide propensity-based features, trinucleotide composition, and electron–ion interaction pseudopotentials, to extract the hidden patterns from the DNA sequence. Afterwards, a stacked framework is formed by integrating a convolutional neural network (CNN) with bidirectional long-short-term memory (LSTM) using multi-view attributes to train the proposed model. The PROCABLES model achieved an accuracy of 0.971 and 0.920 and the MCC 0.940 and 0.840 for the first and second-layer using the ten-fold cross-validation test, respectively. The predicted results anticipate that the proposed PROCABLES protocol outperformed the advanced computational predictors targeting promoters and their types. In summary, this research will provide useful hints for the recognition of large-scale promoters in particular and other DNA problems in general.

1. Introduction

Promoters are specific DNA sequences found upstream of a gene in an organism’s genome. They are essential in initiating gene transcription, which marks the initial phase in the gene expression process [1]. Transcription factors recognize and bind to promoters, which then recruit the enzyme RNA polymerase to the gene’s coding area [2]. A promoter’s fundamental function is to control the commencement of gene transcription [3]. It functions as a binding location for transcription factors and other regulatory proteins, controlling the timing and magnitude of gene expression. Promoters can either boost or repress gene transcription by interacting with these proteins [4]. The efficiency and degree of gene transcription are determined by the strength of a promoter. Strong promoters [5] are connected with increased gene expression, whereas weak promoters [6] result in decreased transcription. The sequence and organization of the promoter elements affect the

affinity of transcription factors and other regulatory proteins, hence impacting total transcriptional activity. By changing the transcripts, promoter mutations can cause tumors, diabetes, Huntington’s disease, and other disorders [7]. Therefore, Promoters were identified from extensive unannotated genetic data using wet lab technological approaches such as mass spectrometry (MS), RNA-seq and X-ray crystallography etc. [8]. Despite their great accuracy, these approaches are expensive, slow and laborious therefore making them unfavorable to analyze and characterized the promoters and their types in large repository of genomic data. [9]. As a result, effective computational algorithms for high-throughput promoter prediction are critical.

Over the past years, remarkable computational-based research has been conducted on promoter’s identification. To the best of our knowledge, numerous sequential-based predictors have been reported in the literature including machine learning (ML) and deep learning (DL) for identifying promoters and their types as illustrated in Fig. 1. For

* Corresponding authors.

E-mail addresses: muhammad.kabir@umt.edu.pk (M. Kabir), mfarif@hbku.edu.qa (M. Arif), talam@hbku.edu.qa (T. Alam).

<https://doi.org/10.1016/j.ymeth.2024.08.005>

Received 25 June 2024; Received in revised form 24 July 2024; Accepted 17 August 2024

Available online 19 August 2024

1046-2023/© 2024 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

example, [10] used the position-correlation scoring matrix (PCSM) technique for targeting promoters from DNA samples and achieved 74.81 % accuracy (ACC). Variable window Z-curve composition (vw Z-curve) [11] and six local DNA structural features [2] and improved the performance of this problem with an ACC of 80.28 %. By incorporating DNA duplex stability into neural networks, two different types of promoters (σ^{54} and σ^{28}) were discovered [12] reached 78.04 % ACC. Later, [13] used PseKNC, an advanced feature in the field of bioinformatics, to identify promoters reached 80.45 % ACC. According to [14], iPromoter-2L were introduced and identified promoters with an ACC of 81.68 %, whereas iPSW(2L)-PseKNC identified promoters with an ACC of 83.13 % [15]. The iPromoter-ET predictor introduced in [6] obtained an ACC of 85.14 % for classification of promoters. A neural network based predictor iEnhancer-ECNN is developed in [16] to further enhance the prediction process and reported an ACC of 85.41 %. Likewise, iPro-GAN is developed which achieved ACC of 93.15 % in [17;18] introduced predictor which is based on attention mechanism achieved ACC of 93.45 %, whereas ‘BERT-Promoter’ identified promoters with an ACC of 85.45 % [19]. According to [20], ‘dPromoter-XGBoost’ was proposed and identified promoters with an ACC of 83.81 %. [21] introduced ‘iPromoter-CLA’ to identify promoters, achieved ACC of 86 %. Another predictor ‘iProTCN’ [22] reported an ACC of 91.86 %. Less study has been done than the previous one for the classifying strength of promoters. [15] introduced that iPSW(2L)-PseKNC reached 71.2 % ACC to classify strength of promoters; [6] iPromoter-ET achieved ACC for the stratification of their strength 72.59 %; [16] conducted study and discovered that the ACC of identifying promoter’s strength reached 73.1 %; [18] proposed that classification of promoter’s strength reached 90.59 % ACC, ‘BERT-Promoter’ identified promoter’s strength with an ACC of 76.92 % [19], ‘dPromoter-XGBoost’ [20] identified promoter’s strength with an ACC of 77 %, ‘iPromoter-CLA’ identify promoter’s strength achieved ACC of 73.46 % [21,22] achieved 84.63 % of ACC. Furthermore, Y Wang et al, developed the tool called PromGER that leverages ensemble-based learning in conjunction with graph embedding to discriminate the eukaryotic regions from DNA sequences [23]. Recently, A Agarwal, proposed a deep-learning based model called DeepPHiC for predicting promoters. The proposed method integrated the epigenomics and genomic attributes to enhance the prediction performance [24].

However, the overall performance of existing computational tools is unsatisfactory and still research gaps available for further improvement. In this study, we developed an improved predictor ‘**PROCABLES**’ (**P**ROMOTERS Classification using **C**NN, **B**iLSTM and **L**STM model**S**) with high efficacy for predicting promoters and their types using sequence

information derived from CKSNAP, PseEIIP, PSTNP, TNC and Word2Vec feature encoding approaches. The developed architecture of DL-based model consists of two stages, in the first-stages it predicts promoters from DNA and in the second stage if it predict promoters then further discriminate their functional types as strong promoters and weak promoters respectively. Extensive experiments show that in both levels ensemble deep learning model (CNN+BiLSTM) model produced best results on the fused features. The schematic layout of *PROCABLES* approach is depicted in Fig. 2. The contribution of present research work can be summarized as follows:

- We designed an intelligent two-layer DL model that predict the promoter region in the first stage and their functional types in second stage respectively.
- We captured the DNA encoded patterns using Word2Vec algorithm.
- We analyzed the visual impact of biological features using t-distributed stochastic neighbor embedding method.
- We enhanced the overall prediction performance of promoters and their functional types (as weak and strong promoters) on both datasets.

2. Materials and methods

2.1. Benchmark dataset

Dataset collection is one of the most vital step in developing a bio-informatics model [25]. We obtained the publicly accessible datasets from previous study [15] to evaluate the designed model’s performance. The benchmark dataset consists of a combined total of 3382 DNA samples, encompassing the sequences of both promoter and non-promoters. Among the promoter sequences, 1591 are strong promoters and 1791 weak promoters. In the benchmark dataset the promoter and non-promoter sequence length was 81 bps. Sequences having a degree of similarity of more than 85 % were excluded using CD-HIT [26,27] in order to prevent the classifier’s divergence. In Table 1 the statistical description of both datasets are given below:

2.2. Feature Engineering

These methods are often used to extract features from DNA sequences. For example, CKSNAP is designed to capture k-spaced nucleotide pairs, PseEIIP focuses on pseudo-emission features, PSTNP considers position-specific trinucleotide propensities, and Word2Vec represents DNA sequences in a continuous vector space. These features

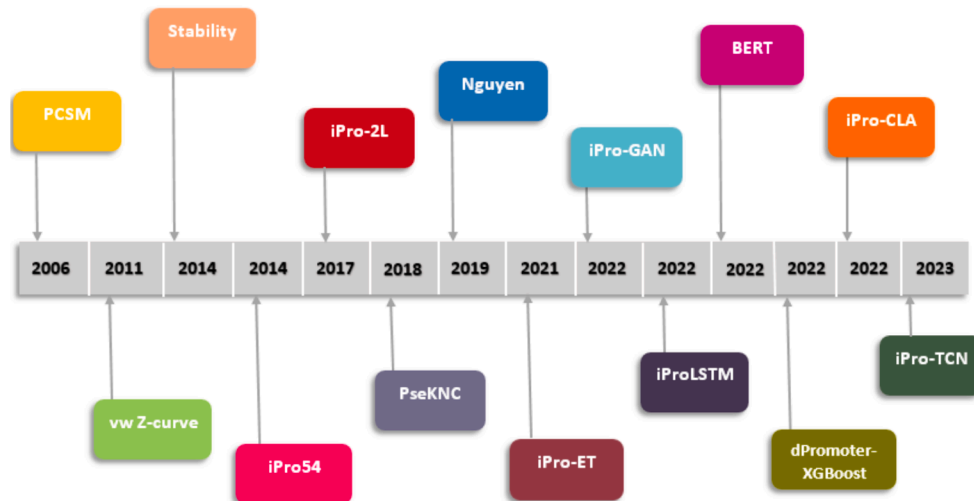


Fig. 1. Timeline of existing predictors.

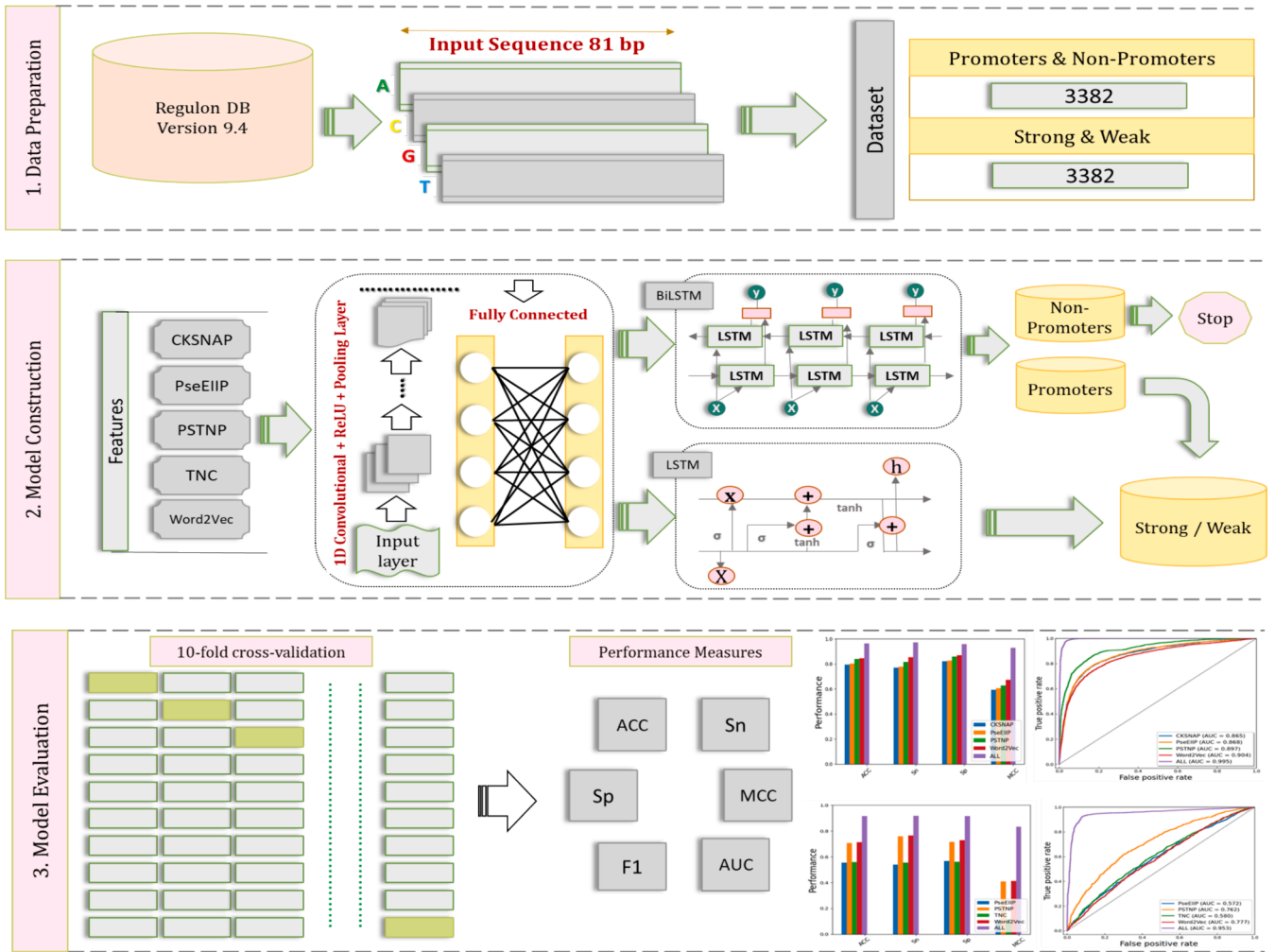


Fig. 2. Flowchart of the proposed PROCABELS.

Table 1

Datasets used in promoter predictors.

Type	Dataset	Samples	CD-HIT threshold
Promoters	S (1st Layer)	3382	0.85
Non-Promoters		3382	
Strong Promoters	S ⁺ (2nd Layer)	1591	
Weak Promoters		1792	

can encode important information of the sequence composition and structure. Moreover, the sequence composition and structure properties are crucial aspects of identifying promoter regions. Promoters often exhibit specific patterns and motifs that can be captured by these features. The rationale behind choosing these features involves a combination of computational efficiency, biological relevance, and the ability to extract the enrich information from raw sequence data.

2.2.1. Composition of *n*-Spaced nucleic-acid pairs

CKSNAP (composition of *n*-Spaced nucleic acid pairs) transforms the biological DNA sample into fixed length statistical features[28] by considering the frequency of *n*-spaced pairs of nucleotides within a sequence. As an illustration, the given DNA sample, 'AAYGYYT', AG and GT, signify nucleotide pairs with two and three spaces between them, respectively. A particular interpretation of KNP frequency is:

$$f(KNP) = \frac{m(KNP)}{N - k - 1}, k \in [0, k_{max}], \quad (1)$$

where (N-k-l) represents the count of KNP along a sequence with a length of N and m(KNP) denotes the total count of KNP within the DNA sample. We maintained the $k_{max} = 2$ that produced a 96D feature vector.

2.2.2. Electron-Ion interaction pseudopotentials

Electron-ion interaction-pseudopotentials (EIIP) is a popular nucleotide-based feature descriptor initially proposed by Nair [29], which has been utilized in various problems such as forecasting cystic-fibrosis transmembrane protein [30], the identification of the enhancer [31–34], and others [35–39]. The EIIP descriptor can mathematically be define as:

$$V = [EIIP_{AAA}f_{AAA}, \dots, EIIP_{TTT}f_{TTT}] \quad (2)$$

where f_{xyz} denotes the normalized frequency and $EIIP_{ABC}$ denotes statistical value of represents the trinucleotide's. The size of vector V is clearly 64D.

2.2.3. Position-Specific trinucleotide propensity

PSTNP stands for position-specific trinucleotide-propensity is a well-known feature representation that has the ability to retrieve DNA sequence location data. The working principle of PSTNP is to initially consider the probability or occurrence of 3 consecutive nucleotide pairs across the entire sequence called 3-mer approach. The PSTNP profile P is then created by subtracting two matrices, and it may be utilized to represent each DNA sequence as a matrix with a length of (L-2) [31]. Total trinucleotide is 64 including AAA, AAC, AAG, ..., TTT. Therefore,

the following 64×79 matrix may be used to describe the trinucleotide position specificity for an 81-bp sample:

$$Y = \begin{bmatrix} Y_{1,1} & Y_{1,2} & \cdots & Y_{1,79} \\ Y_{2,1} & Y_{2,2} & \cdots & Y_{2,79} \\ \vdots & \vdots & \cdots & \vdots \\ Y_{64,1} & Y_{64,2} & \cdots & Y_{64,79} \end{bmatrix} \quad (3)$$

where the variable:

$$Y_{ij} = P^+(3mer_i|j) - P^-(3mer_i|j) \quad (4)$$

$$S = [\varphi_1, \varphi_2, \dots, \varphi_u, \dots, \varphi_{79}]^T \quad (5)$$

Where the operator T denotes the transpose and φ_u was defined as follows:

$$\varphi_u = \begin{bmatrix} Y_{1,u} & \text{when } N_u N_{u+1} N_{u+2} = AAA \\ Y_{2,u} & \text{when } N_u N_{u+1} N_{u+2} = AAC \\ Y_{3,u} & \text{when } N_u N_{u+1} N_{u+2} = AAG \\ \vdots & \vdots \\ Y_{64,u} & \text{when } N_u N_{u+1} N_{u+2} = TTT \end{bmatrix} \quad (6)$$

Where $1 \leq u \leq 79$.

2.2.4. Trinucleotide composition

Trinucleotide Composition (TNC) feature extraction approach has widely been adopted [40–42] to compute the frequency of three nucleotides. The feature vector created by TNC can mathematically defined as:

$$V = [f(AAA)f(AAC)f(AAG)f(AAT)\dots f(TTT)]^T \quad (7)$$

$$V = [f_1 f_2 f_3 \dots f_{64}]^T \quad (8)$$

where the rates of occurrence for AAA, AAC, AAG, AAT, and TTT are represented by $f(TTT)$, $f(GAA)$, $f(GAT)$, $f(ATG)$, and $f(GGG)$, respectively. According to this, the feature vector for TNC will be $4 \times 4 \times 4 = 64$ -D in size [43]. The following equation can be used to express the simplified version of 4^k factors [44]:

$$V = [f_1^k f_2^k f_3^k \dots f_{4^k}^k]^T \quad (9)$$

2.2.5. Word2Vec

Based on the idea that phrases with similar significance in a certain context display close distances, Word2Vec analyses [45,46] and transforms the conceptual content of phrases in texts. The Mikolov-proposed Word2Vec learning algorithms CBOW and Skip-gram. There are input, projection, and output layers in both learning methods, but having different methods for determining the outcome.

$$V_n = \{V_{[x-2]}, V, \dots, V_{[x+1]}, V_{[x+2]}\} \quad (10)$$

Above Eq. (10) is passed as parameters to the input layer, where V_n stands for words. The projection plane preserves the total of many vectors, which is equivalent to a collection of multivariate vectors [47]. The element from which the projection layer's trajectory outputs are produced is known as the output layer.

2.3. Classification algorithms

Different categorization techniques have been used in the current investigation to precisely identify promoters.

2.3.1. Convolutional neural network

We used 2D CNN, included some layers such as input layer which is

1D vector of proposed model and length of vector 1×100 , next layer is convolutional layer. Convolution kernels are built using a 1D convolutional layer (for instance, secular convolution), and features stored in the 1D input vector are then extracted. The convolutional layer advances to the input while using a sliding window to convert the values into representative values. Through the acquisition of useful features utilizing only a portion of the input data, this approach aids in the preservation of the spatial association between the numerical data in the features.

2.3.2. Long short-term memory

Sequence prediction and classification use a recurrent neural network (RNN) called long short-term memory (LSTM), which can learn long-term dependencies in a sequence. It consists of a number of memory cells, each of which has three gates—input, output, and forget. In this situation, the LSTM will selectively recall and forget items [48]. It can learn and recognize the lengthy sequence; the current state is calculated by using Eq:

$$W_t = f(W_{t-1}, X_t) \quad (11)$$

Where X_t is the input state, W_t is the current state, and W_{t-1} is the prior state.

2.3.3. Bidirectional LSTM

The model now has to learn the links between past and future words when processing large data sequences. This problem was addressed with the development of the bidirectional network. We can use a bidirectional network with long-short-term memory (LSTM). A recurrent neural network called Bidirectional LSTM (BiLSTM) is mostly utilized for natural language processing. It can use data from both sides, and input flows in both directions, unlike a typical LSTM. It works well for mimicking word and sentence connections in both directions of the sequence.

2.4. Data and code availability

The dataset and other source code materials are provided at <https://github.com/mdkabr1/PROCABLES>.

2.5. Architecture of proposed PROCABLES

Several fundamental deep models may experience over-fitting and ensemble diversity problems when dealing with small sample sizes [49,50]. These techniques can teach classifiers more pertinent high-level characteristics. It has emerged as the industry standard classifier in a number of areas, including the classification of cancer subtypes, face anti-spoofing, hyperspectral imaging, radar high resolution range profile recognition, and self-interacting proteins. Bioinformatics data is often heterogeneous, coming from different experimental techniques or sources. Feature fusion techniques can help integrate diverse types of information, making the model more adaptable to varying data modalities. This aid our model in gaining insights into the factors influencing the prediction of promoters and non-promoters. PROCABLES is an ensemble-based method that uses feature fusion approaches CKSNAP, PseEIP, PSTNP, TNC and Word2vec with CNN [51], LSTM [52], and BiLSTM [53] layers rather than specific neural network topologies for feature learning in each layer. Due to its deep-type design, the deep model's technique is dependable and appropriate for training with small amounts of data. PROCABLES is better at tweaking the hyper parameter than DNN. The PROCABLES classifier has a higher ability for discrimination than other classification algorithms due to its excellent learning potential. To carry out this experiment, we used Python 3.10.10, TensorFlow 2.10.0, and Keras 2.10.0. The hyper parameters in this work as provided Supplementary file Table S1. To identify promoters, we used 2 layers of CNN+BiLSTM and dense layer as well while to classify strength

of promoters, 1 layer of CNN+LSTM and dense layer is used and set epoch size to 30, batch-size to 128 and learning rate to 0.1 for both layers.

2.6. Performance metrics and evaluation criteria

Various evaluation metrics are employed to examine the effectiveness of predictive models [54,55]. These metrics provide quantitative measures of how well the model performs on the validation sets during each iteration of the cross-validation (CV) process [56–58]. We used 10-fold CV test method to build the prediction model [59]. Some commonly used performance evaluation metrics used in this study are specificity (Sp), accuracy (ACC), sensitivity (Sn), receiver operating curve-area under the curve (AUC -ROC), and Mathew's correlation coefficient (MCC) [60–63].

$$ACC = \frac{TP + TN}{TP + FP + FN + TN} \quad (12)$$

$$Sn = \frac{TP}{TP + FN} \quad (13)$$

$$Sp = \frac{TN}{FP + TN} \quad (14)$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{[TP + FP][TP + FN][TN + FP][TN + FN]}} \quad (15)$$

In this context TP represents true prediction of promoters, TN denotes true predictions of non-promoters, FP represents false prediction of non-promoters as promoters and FN denotes the false prediction of promoters as non-promoters.

3. Result and discussion

3.1. Classification of promoters

Sequence information on many aspects can be obtained using various feature representation techniques. Different features may capture distinct aspects of the regulatory regions. Feature fusion allows the model to leverage the complementary information provided by each feature, potentially improving the overall predictive performance, enables the model to learn and adapt to intricate relationships between different features, providing a more nuanced understanding of the regulatory mechanisms. It can also be employed as a form of dimensionality reduction. Instead of dealing with a large number of individual features, combining them into a smaller set of fused features can simplify the model and reduce the risk of overfitting. For the first-layer of the present

study, we used four distinct feature representation approaches involves CKSNAP, PseEIIP, PSTNP, and Word2Vec reached ACC of 0.795, 0.802, 0.839 and 0.844. With feature fusion we achieved 0.971 of ACC presented graphically in Fig. 3, it is analyzed that achieved results with feature fusion are higher than individual features as shown in Table 2. We then fed them to the combination of deep learning classifiers CNN+BiLSTM via 10-fold CV method because the trained model's performance is more uniform as the number of fold increases and evaluate the performance of model reached 0.995 AUC. Further, we analyzed and compare the prediction efficacy of the proposed model with existing algorithms such as CNN, DNN, GRU, LSTM, MLP, RF, RNN and SVM on benchmark dataset, illustrated in Fig. 4. It is clear from Table 3 that among all our fusion classifier CNN+BiLSTM achieved highest results with ACC 0.971 and MCC of 0.94 as shown in Table 3, respectively. We also showed that these deep learning models outperform general classifiers in terms of prediction accuracy and learning capacity. Our model uses a deep learning framework in comparison to a single model, which has the advantages of great learning capacity and quick speed.

3.2. Stratification of Promoter's strength

Various feature representation approaches may be used to get sequence information on a wide range of factors. A single feature can't gather correct data, hence multi-feature is being employed more and more in bioinformatics. For the second-layer of the present study, we used four distinct feature representation approaches involves TNC, PseEIIP, PSTNP, and Word2Vec reached ACC of 0.558, 0.556, 0.707 and 0.714. With feature fusion we achieved 0.920 of ACC illustrated in Fig. 5, it is analyzed that achieved results with feature fusion are higher than individual features as shown in Table 4. We then fed them to the combination of deep learning classifiers CNN+LSTM via 10-fold CV method because the trained model's performance is more uniform as the number of fold increases and then evaluate the performance of model reached 0.953 AUC. Furthermore, we used a benchmark dataset to compare the performance of our proposed fusion classifiers to existing classifiers for stratification of promoter's strength such as CNN, DNN, GRU, LSTM, MLP, RF, RNN, and SVM depicted in Fig. 6. As shown in Table 5, our fusion classifier CNN+LSTM achieved the best results with an ACC of 0.92 and an MCC of 0.84, respectively. We employ the edge of convolution dimension reduction in CNN to learn more effective features in comparison to the standard deep model, and we employ the recurrent structure of GRU to selectively recall relevant information and discard less significant information.

3.3. Comparison with existing models

We compared the success rates of PROCABLES model to the existing

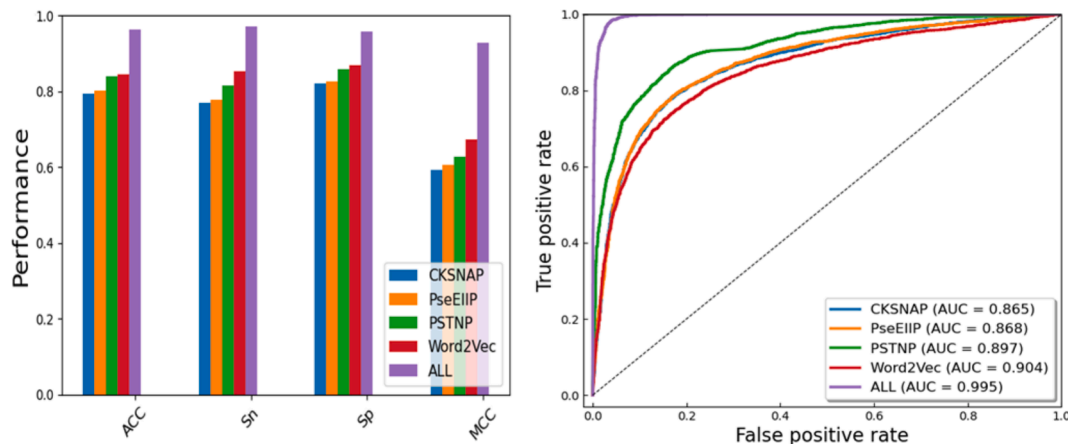


Fig. 3. Performance Analysis of different Features using benchmark dataset at first-layer.

Table 2
Comparison of feature-encoding first-layer via 10-fold CV.

Layer	Approach	ACC	MCC	Sn	Sp	AUC	Precision	F1	TP	FN	TN	FN
Layer 1	CKSNAP	0.795	0.592	0.769	0.821	0.865	0.782	0.771	2575	719	2663	807
	PseEIIP	0.802	0.606	0.779	0.826	0.868	0.769	0.768	2574	748	2634	808
	PSTNP	0.839	0.627	0.816	0.859	0.897	0.797	0.267	2670	681	2701	712
	Word2Vec	0.844	0.673	0.854	0.869	0.904	0.762	0.248	2488	654	2728	894
	ALL	0.971	0.940	0.985	0.956	0.995	0.965	0.972	3314	118	3264	68

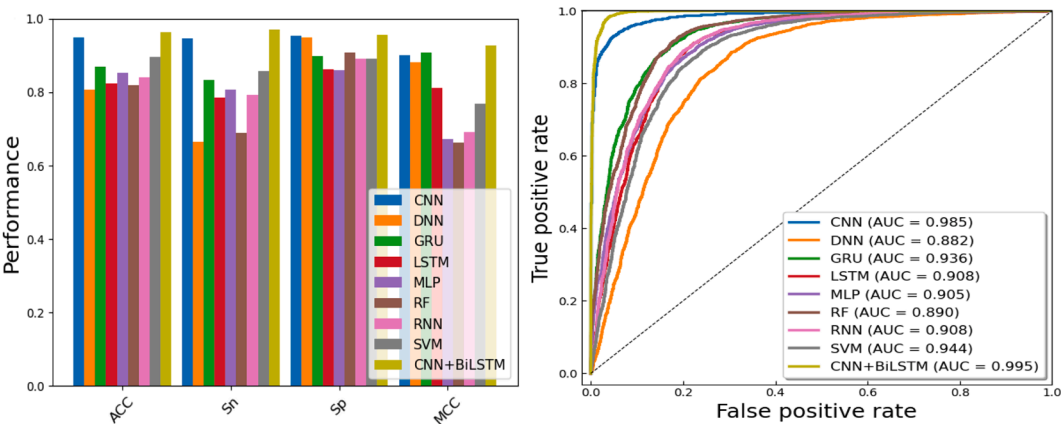


Fig. 4. Performance Analysis of classification algorithms via 10-fold CV validation for first-layer.

Table 3
Comparison of classification algorithms via 10-fold CV on a benchmark dataset for first-layer.

Layer	Method	ACC	MCC	Sn	Sp	AUC
Layer 1	CNN	0.950	0.902	0.947	0.954	0.985
	DNN	0.807	0.645	0.665	0.950	0.882
	GRU	0.870	0.746	0.833	0.908	0.936
	LSTM	0.824	0.811	0.786	0.863	0.908
	MLP	0.852	0.673	0.808	0.859	0.905
	RF	0.818	0.663	0.689	0.907	0.890
	RNN	0.842	0.692	0.792	0.892	0.908
	SVM	0.895	0.769	0.858	0.892	0.944
	CNN+BiLSTM	0.971	0.940	0.985	0.956	0.995

approaches, as illustrated in Fig. 7. When the new technique and current models were compared, the new method produced superior results than the existing predictors. The comparison has been drawn with 11 predictors to identify promoters, including PCSF [10], vw Z-curve [11], Stability [12], iPro54 [13], iPromoter-2L, iPSW(2L)PseKNC [15,6]

iPromoters-ET and combination of continuous FastText N-grams [16], iPro-GAN [17], LSTM based attention mechanism [18] and iPro-TCN [64].

The outcomes of the suggested method are better than those of earlier techniques as shown in Table 6. The progress rate of ACC, MCC, Sn and Sp are 3.66 %, 0.08 %, 5.08 % and 2.25 %, when compared LSTM based attention mechanism model for classification of promoters. Meanwhile, for the stratification of promoter’s strength, comparison made with same predictor and improved 1.45 % ACC, 0.03 of MCC, 2.55 % of Sn and 0.44 % of Sp, respectively. Hence, proposed model outperforms for the prediction of promoters and their strength.

3.4. Analysis of the feature descriptors

In this subsection, we visualize the feature extraction techniques by using two-dimensional scatter plotting algorithm called tSNE [51] to analyze the model interpretability. Figs. 8 and 9 represents features visualization of both datasets depicting our model achievement over individual and hybrid sequential features. We design a two-dimensional

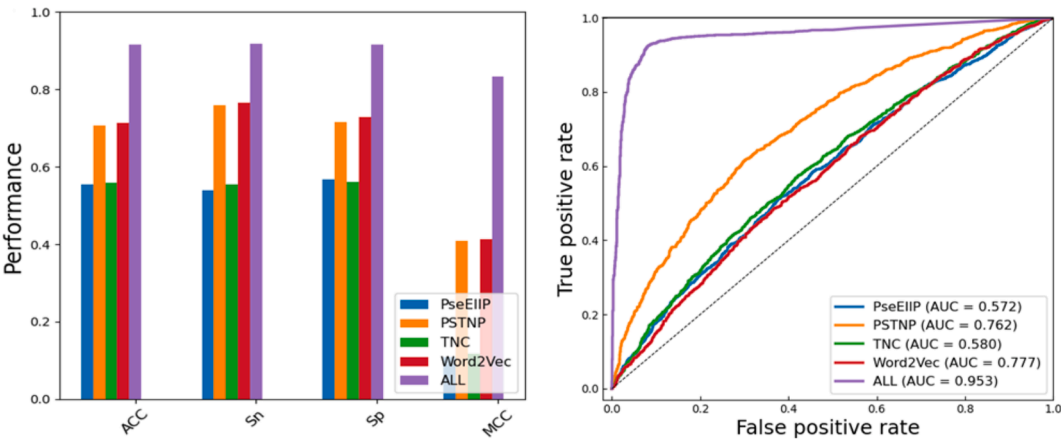


Fig. 5. Performance Analysis of different Features using benchmark dataset at second-layer.

Table 4
Comparison of feature-encoding for second-layer via 10-fold CV.

Layer	Approach	ACC	MCC	Sn	Sp	AUC	Precision	F1	TP	FN	TN	FN
Layer 2	PseEIIP	0.555	0.109	0.540	0.567	0.572	0.561	0.567	998	597	1094	693
	PSTNP	0.707	0.408	0.759	0.715	0.762	0.625	0.605	977	763	928	714
	TNC	0.558	0.117	0.555	0.562	0.580	0.545	0.548	970	825	866	721
	Word2Vec	0.713	0.413	0.766	0.729	0.777	0.560	0.552	923	719	972	768
	ALL	0.920	0.841	0.928	0.913	0.953	0.928	0.926	1566	126	1565	125

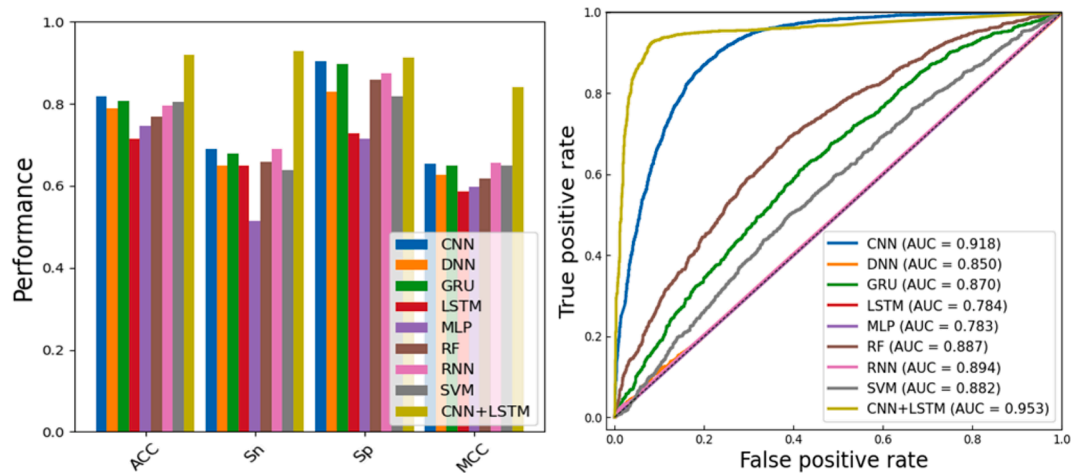


Fig. 6. Performance Analysis of classification algorithms via 10-fold CV validation for second-layer.

Table 5
Comparison of classifiers via 10-fold CV on a benchmark dataset for second-layer.

Layer	Method	ACC	MCC	Sn	Sp	AUC
Layer 2	CNN	0.818	0.653	0.689	0.904	0.918
	DNN	0.789	0.626	0.649	0.829	0.850
	GRU	0.807	0.650	0.678	0.898	0.870
	LSTM	0.714	0.587	0.650	0.729	0.784
	MLP	0.747	0.598	0.514	0.715	0.783
	RF	0.769	0.618	0.659	0.858	0.887
	RNN	0.797	0.656	0.689	0.874	0.894
	SVM	0.805	0.649	0.638	0.819	0.882
	CNN+LSTM	0.920	0.841	0.928	0.913	0.953

strategy with the data [65]. A sample is represented by each point. Green and red colors are used to denote promoters and non-promoters, as well as strong and weak samples. Five different features i.e., CKSNAP, PseEIIP, PSTNP, Word2Vec and ALL are used as DNA encoder for identification of promoters and non-promoter’s samples in the first-layer and strong and weak promoters in the second-layer respectively. It can be analyzed from the given representations that; the combined features are more effective in discriminative process of both layers. Among the individual features CKSNAP performance is weak as compared to others whereas Word2Vec provided higher performance from all the others.

4. Conclusion

The prediction of DNA promoters and their strength, play significant role in analyzing gene regulation. In this research, we proposed a two-

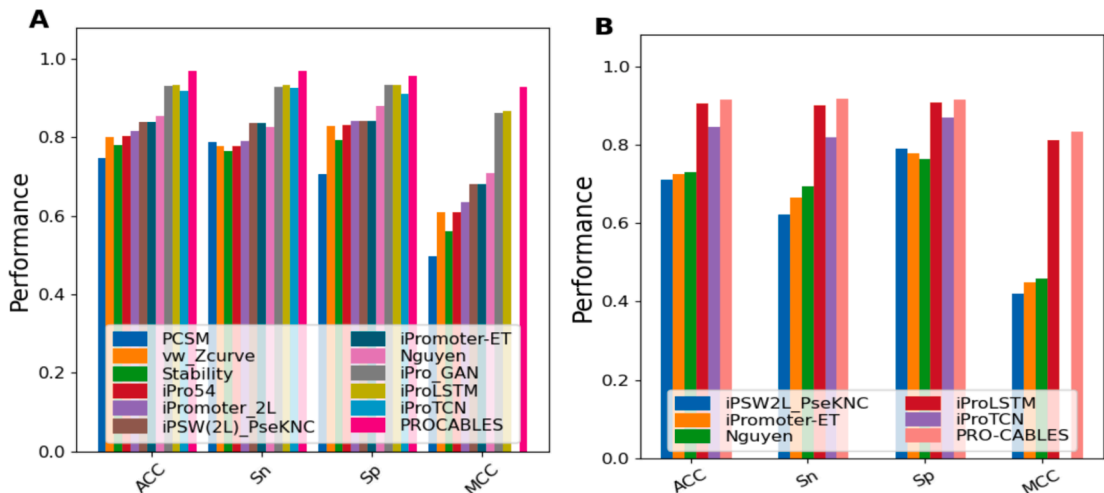


Fig. 7. Comparison of the proposed method with the current method at first-layer (A) and second-layer (B).

Table 6
Analogy of the proposed model with the current models.

Layers	Methods	ACC	MCC	Sn	Sp
Layer 1	PCSM	0.748	0.490	0.789	0.707
	vw Z-curve	0.802	0.600	0.777	0.828
	Stability	0.780	0.560	0.766	0.794
	iPro54	0.804	0.610	0.777	0.831
	iPromoter-2L	0.816	0.630	0.792	0.841
	iPSW(2L)-PseKNC	0.840	0.681	0.837	0.843
	iPromoter-ET	0.851	0.703	0.842	0.860
	Nguyen et al.	0.854	0.700	0.827	0.880
	iPro-GAN	0.931	0.860	0.929	0.933
	iProLSTM	0.934	0.860	0.934	0.934
	BERT-Promoter	0.854	—	0.843	0.865
	dPromoter-XGBoost	0.838	—	0.857	0.819
	iPromoter-CLA	0.860	0.721	0.868	0.851
	iProTCN	0.918	—	0.927	0.910
	PROCABLES	0.971	0.940	0.985	0.956
Layer 2	iPSW(2L)-PseKNC	0.712	0.420	0.622	0.791
	iPromoter-ET	0.725	0.449	0.666	0.779
	Nguyen et al.	0.731	0.460	0.694	0.764
	iProLSTM	0.905	0.810	0.902	0.909
	BERT-Promoter	0.769	—	0.708	0.816
	dPromoter-XGBoost	0.770	—	0.806	0.729
	iPromoter-CLA	0.734	0.470	0.776	0.687
	iProTCN	0.846	—	0.819	0.870
	PROCABLES	0.920	0.840	0.928	0.913

layer predictor named *PROCABLE* that uses CKSNAP, PseEIIP, PSTNP and Word2Vec as DNA encoding methods and CNN+BiLSTM algorithm as base-learning model to identify promoters and their functional types. Furthermore, we made a feature comparison between single and fused features, classifiers-based comparison between ML and DL models on both layer datasets respectively. The developed protocol surpassed the recent state-of-the-art two-phase promoter predictors. In near future, we will make an effort to establish a publicly accessible webserver for scientific purpose. We are expecting this research will significantly contribute in detecting large-scale promoters and pharmaceutical

companies for the drug designing. It is paramount to acknowledge the importance of the dataset utilized in this study. The dataset primarily focuses on prokaryotic promoters, which may not fully capture the complexity of eukaryotic promoters relevant to human disorders. As a result, the findings of this study may be limited in their applicability to understanding promoter mutations in eukaryotic systems. Future research efforts should consider incorporating datasets that encompass a broader range of promoter types to provide a more comprehensive understanding of promoter function and regulation in human disorders.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and material

Data used in this article is publicly available.

Author contributions

Aqsa Amjad: Data Collection, Conceptualization, Methodology, Software, Experiments Validation, Formal analysis, Investigation, Writing – original draft. Saeed Ahmad: Validation, Investigation, Resources, Visualization, Project administration, Supervision. Muhammad Kabir: Formal Analysis, Conceptualization, Methodology, Validation, Experiments, Supervision. Muhammad Arif: Formal Analysis, Validation, Investigation, Resources, Visualization, Writing – review & editing, Tanvir Alam: Formal Analysis, Validation, Investigation, Resources, Visualization, Supervision, Funding, Writing – review & editing.

Funding

The open access publication of this work was supported by the

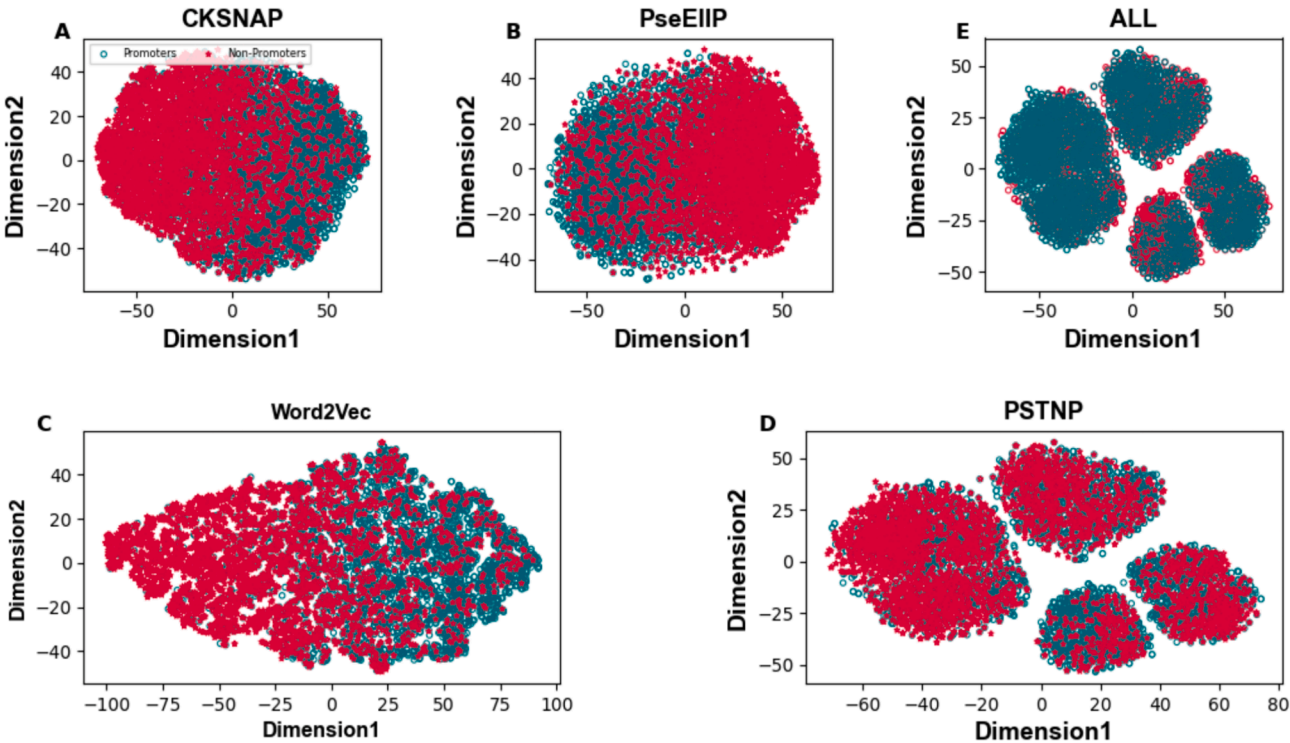


Fig. 8. T-sne analysis of promoters (green) and non-promoters (red) dna sequence for the first-layer (a, b, c, d, e): cksnap (a), pseeiip (b), word2vec (c), pstnp (d) and E (ALL).

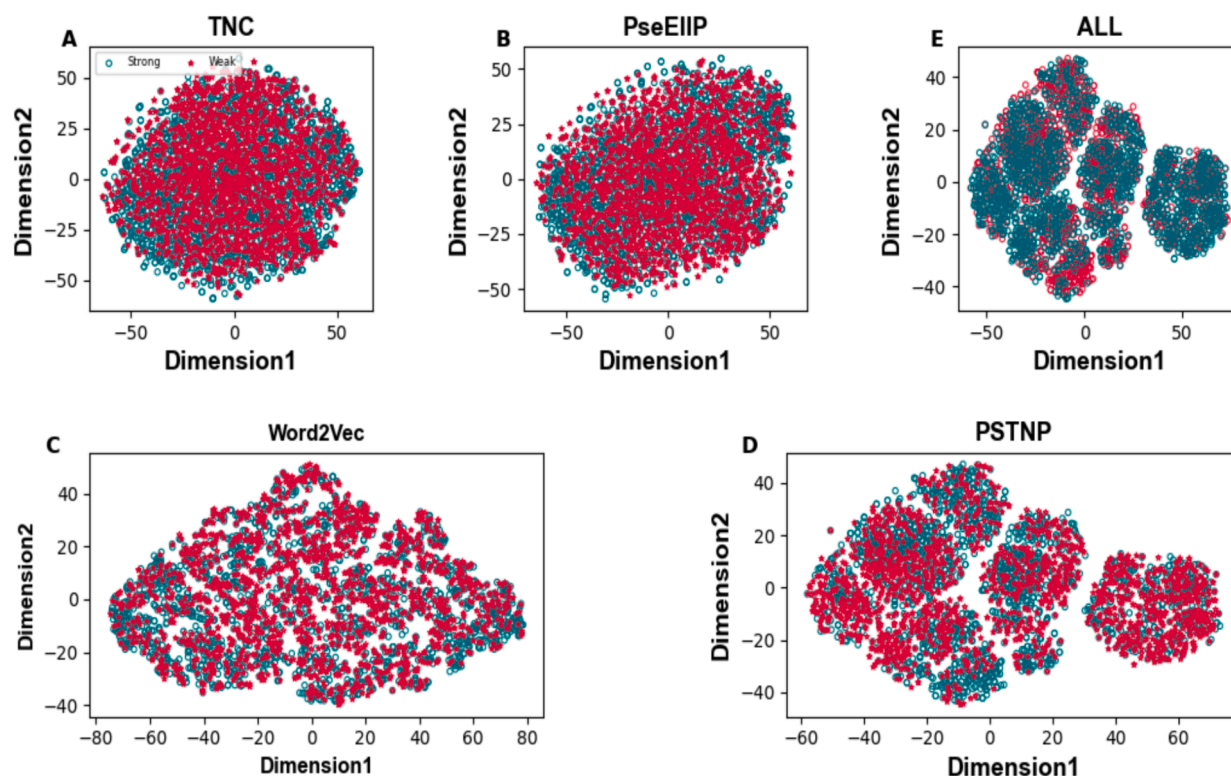


Fig. 9. Analysis of strong (via green) and weak (via red) promoters for the second-layer (A, B, C, D, E): CKSNAP (A), PseEIIP (B), Word2Vec (C), PSTNP (D) and E(ALL).

College of Science and Engineering, Hamad Bin Khalifa University, Doha 34110, Qatar.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ymeth.2024.08.005>.

References

- [1] F.S. Collins, L. Fink, The Human genome project, *Alcohol Health Res World* 19 (3) (1995) 190–195.
- [2] H. Lin, et al., Identifying Sigma70 Promoters with Novel Pseudo Nucleotide Composition, *IEEE/ACM Trans Comput Biol Bioinform* 16 (4) (2019) 1316–1321.
- [3] V. Haberle, A. Stark, Eukaryotic core promoters and the functional basis of transcription initiation, *Nat Rev Mol Cell Biol* 19 (10) (2018) 621–637.
- [4] M. Lonnie, et al., Protein for life: review of optimal protein intake, sustainable dietary sources and the effect on appetite in ageing adults, *Nutrients* 10 (3) (2018).
- [5] Schlabach, M.R., et al., *Synthetic design of strong promoters*. 2010. 107(6): p. 2538–2543.
- [6] Y. Liang, et al., iPromoter-ET: Identifying promoters and their strength by extremely randomized trees-based feature selection, *Anal Biochem* 630 (2021) 114335.
- [7] S. Döhr, et al., Linking disease-associated genes to regulatory networks via promoter organization, *Nucleic Acids Res* 33 (3) (2005) 864–872.
- [8] Y. Zhang, et al., Model-based analysis of ChIP-Seq (MACS), *Genome Biology* 9 (9) (2008) R137.
- [9] M.G. Reese, Application of a time-delay neural network to promoter annotation in the *Drosophila melanogaster* genome, *Comput Chem* 26 (1) (2001) 51–56.
- [10] Q.Z. Li, H. Lin, The recognition and prediction of sigma70 promoters in *Escherichia coli* K-12, *J Theor Biol* 242 (1) (2006) 135–141.
- [11] K. Song, Recognition of prokaryotic promoters based on a novel variable-window Z-curve method, *Nucleic Acids Res* 40 (3) (2012) 963–971.
- [12] S. de Avila e Silva, et al., DNA duplex stability as discriminative characteristic for *Escherichia coli* σ 54- and σ 28- dependent promoter sequences, *Biologicals* (2013) 42.
- [13] H. Lin, et al., iPro54-PseKNC: a sequence-based predictor for identifying sigma-54 promoters in prokaryote with pseudo k-tuple nucleotide composition, *Nucleic Acids Res* 42 (21) (2014) 12961–12972.
- [14] B. Liu, et al., iPromoter-2L: a two-layer predictor for identifying promoters and their types by multi-window-based PseKNC, *Bioinformatics* 34 (1) (2018) 33–40.
- [15] X. Xiao, et al., iPSW(2L)-PseKNC: A two-layer predictor for identifying promoters and their strength by hybrid features via pseudo K-tuple nucleotide composition, *Genomics* 111 (6) (2019) 1785–1793.
- [16] Q.H. Nguyen, et al., iEnhancer-ECNN: identifying enhancers and their strength using ensembles of convolutional neural networks, *BMC Genomics* 20 (9) (2019) 1–10.
- [17] H. Qiao, et al., iPro-GAN: A novel model based on generative adversarial learning for identifying promoters and their strength, *Comput Methods Programs Biomed* 215 (2022) 106625.
- [18] Q. Li, et al., Identification and classification of promoters using the attention mechanism based on long short-term memory, *Front Comput Sci* 16 (4) (2022) 164348.
- [19] N.Q.K. Le, et al., BERT-Promoter: an improved sequence-based predictor of DNA promoter using BERT pre-trained model and SHAP feature selection, *Comput Biol Chem* 99 (2022) 107732.
- [20] H. Li, et al., dPromoter-XGBoost: Detecting promoters and strength by combining multiple descriptors and feature selection using XGBoost, *Methods* 204 (2022) 215–222.
- [21] Z.-M. Zhang, et al., iPromoter-CLA: Identifying promoters and their strength by deep capsule networks with bidirectional long short-term memory, *Comput Methods and Progra Biomed* 226 (2022) 107087.
- [22] A. Raza, M. Tahir, W. Alam, iPro-TCN: prediction of DNA promoters recognition and their strength using temporal convolutional network, *IEEE Access* (2023) 1.
- [23] Y. Wang, et al., PromGER: promoter prediction based on graph embedding and ensemble learning for eukaryotic sequence, *Genes* 14 (7) (2023) 1441.

- [24] A. Agarwal, L. Chen, DeepPHiC: predicting promoter-centered chromatin interactions using a novel deep learning approach, *Bioinformatics* 39 (1) (2023) p. btac801.
- [25] M. Arif, et al., StackACPred: prediction of anticancer peptides by integrating optimized multiple feature descriptors with stacked ensemble approach, *Chemomet Intell Labor Syst* 220 (2022) 104458.
- [26] W. Li, A. Godzik, Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences, *Bioinformatics* 22 (13) (2006) 1658–1659.
- [27] L. Fu, et al., CD-HIT: accelerated for clustering the next-generation sequencing data, *Bioinformatics* 28 (23) (2012) 3150–3152.
- [28] B. Manavalan, et al., Computational prediction of species-specific yeast DNA replication origin via iterative feature representation, *Brief Bioinform* 22 (4) (2021).
- [29] A.S. Nair, S.P. Sreenadhan, A coding measure scheme employing electron-ion interaction pseudopotential (EIIP), *Bioinformation* 1 (6) (2006) 197–202.
- [30] Rao, K. and M.N.s. Swamy, *Analysis of Genomics and Proteomics Using DSP Techniques*. Circuits and Systems I: Regular Papers, IEEE Transactions on, 2008. 55: p. 370–378.
- [31] W. He, C. Jia, EnhancerPred2.0: predicting enhancers and their strength based on position-specific trinucleotide propensity and electron–ion interaction potential feature selection, *Mol BioSyst* 13 (4) (2017) 767–774.
- [32] S. Blinka, et al., Identification of transcribed enhancers by genome-wide chromatin immunoprecipitation sequencing, *Methods Mol Biol* 1468 (2017) 91–109.
- [33] C.J. Coppola, R.C. Ramaker, E.M. Mendenhall, Identification and function of enhancers in the human genome, *Hum Mol Genet* 25 (R2) (2016). R190–R197.
- [34] K. Niu, et al., iEnhancer-EBLSTM: identifying enhancers and strengths by ensembles of bidirectional long short-term, *Memory* 12 (2021).
- [35] S.S. Sahu, G. Panda, Efficient localization of hot spots in proteins using a novel S-transform based filtering approach, *IEEE/ACM Trans Comput Biol Bioinform* 8 (5) (2011) 1235–1246.
- [36] Y. Park, et al., Identification of human gene research articles with wrongly identified nucleotide sequences, *Life Sci Alliance* 5 (4) (2022).
- [37] Y. Zhang, D. Yin, H. Song, Genome-wide identification and characterization of gene families in arachis, *Methods and Strategies*. 11 (2020).
- [38] T.P. Frazier, et al., Identification, characterization, and gene expression analysis of nucleotide binding site (NB)-type resistance gene homologues in switchgrass, *BMC Genomics* 17 (1) (2016) 892.
- [39] S.E. Antonarakis, History of the methodology of disease gene identification, *Am J Med Genet A* 185 (11) (2021) 3266–3275.
- [40] Y. Xu, et al., iNitro-Tyr: prediction of nitrotyrosine sites in proteins with general pseudo amino acid composition, *PLoS One* 9 (8) (2014) e105018.
- [41] W.R. Qiu, X. Xiao, K.C. Chou, iRSpot-TNCPseAAC: identify recombination spots with trinucleotide composition and pseudo amino acid components, *Int J Mol Sci* 15 (2) (2014) 1746–1766.
- [42] M. Kabir, M. Hayat, iRSpot-GAEnsC: identifying recombination spots via ensemble classifier and extending the concept of Chou's PseAAC to formulate DNA samples, *Mol Genet Genom* 291 (2016) 285–296.
- [43] M. Tahir, M. Hayat, M. Kabir, Sequence based predictor for discrimination of enhancer and their types by applying general form of Chou's trinucleotide composition, *Comput Methods Progr Biomed* 146 (2017) 69–75.
- [44] M. Kabir, et al., iTIS-PseKNC: Identification of Translation Initiation Site in human genes using pseudo k-tuple nucleotides composition, *Comput Biol Med* 66 (2015) 252–257.
- [45] Q. Chen, M. Sokolova, Specialists, scientists, and sentiments: word2vec and doc2vec in analysis of scientific and medical texts, *SN Comput Sci* 2 (5) (2021) 414.
- [46] S. Al-Saqqa, A. Awajan, The use of word2vec model in sentiment analysis, *A Survey*. (2019).
- [47] N. Mengual-Macennle, et al., Multivariate analysis in thoracic research, *J Thorac Dis* 7 (3) (2015) E2–E6.
- [48] Sakalle, A., et al., *A LSTM based deep learning network for recognizing emotions using wireless brainwave driven system*. 2021. 173(C %J Expert Syst. Appl.): p. 19.
- [49] I.H. Sarker, Deep learning: a comprehensive overview on techniques, taxonomy, applications and research directions, *SN Comput Sci* 2 (6) (2021) 420.
- [50] Arif, M., et al., *DeepCPPred: a deep learning framework for the discrimination of cell-penetrating peptides and their uptake efficiencies*. IEEE/ACM Transactions on Computational Biology Bioinformatics, 2021.
- [51] L. Alzubaidi, et al., Review of deep learning: concepts, CNN architectures, challenges, applications, future directions, *J Big Data* 8 (1) (2021) 53.
- [52] S. Hochreiter, J. Schmidhuber, Long short-term memory, *Neural Comput* 9 (1997) 1735–1780.
- [53] Islam, M.S., A Deep Recurrent Neural Network with BiLSTM model for Sentiment Classification. 2018.
- [54] S. Ahmed, et al., An integrated feature selection algorithm for cancer classification using gene expression data, *Combinat Chem High Throughput Screen* 21 (9) (2018) 631–645.
- [55] Arif, M., et al., DeepCPPred: a deep learning framework for the discrimination of cell-penetrating peptides and their uptake efficiencies. 2021. 19(5): p. 2749–2759.
- [56] M. Kabir, et al., Prediction of membrane protein types by exploring local discriminative information from evolutionary profiles, *Anal Biochem* 564 (2019) 123–132.
- [57] Kabir, M., et al., Improving prediction of extracellular matrix proteins using evolutionary information via a grey system model and asymmetric under-sampling technique. 2018. 174: p. 22–32.
- [58] M. Kabir, et al., Prediction of membrane protein types by exploring local discriminative information from evolutionary profiles, *Anal Biochem* 564 (2018) 123–132.
- [59] P. Charoenkwan, et al., iBitter-fuse: a novel sequence-based bitter peptide predictor by fusing multi-view features, *Int J Mol Sci* 22 (16) (2021) 8958.
- [60] P. Charoenkwan, et al., In silico approaches for the prediction and analysis of antiviral peptides: a review, *Curr Pharm Design* 27 (18) (2021) 2180–2188.
- [61] S. Ahmad, et al., SCORPION is a stacking-based ensemble learning framework for accurate prediction of phage virion proteins, *Sci Rep* 12 (1) (2022) 4106.
- [62] P. Charoenkwan, et al., SAPPHERE: A stacking-based ensemble learning framework for accurate prediction of thermophilic proteins, *Comput Biol Med* 146 (2022) 105704.
- [63] P. Charoenkwan, et al., A novel sequence-based predictor for identifying and characterizing thermophilic proteins using estimated propensity scores of dipeptides, *Sci Rep* 11 (1) (2021) 23782.
- [64] A. Raza, et al., iPro-TCN: prediction of DNA promoters recognition and their strength using temporal convolutional network, *IEEE Access* 11 (2023) 66113–66121.
- [65] S. Ahmed, et al., DeepPPSite: a deep learning-based model for analysis and prediction of phosphorylation sites using efficient sequence information, *Anal Biochem* 612 (2021) 113955.